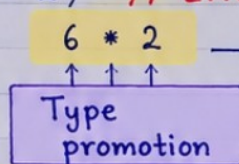


Type Promotion in Expression

- Type promotion means when we perform an operation (like +, -, *, /, %, etc.) on small data types (like byte, short, char), they are **automatically converted to int before the operation is done** (calculated).

① Example : byte operation

```
byte b = 5;
byte a = 6 * 2; // Error
```



→ 6 is a byte
→ 2 is an int
→ (6 * 2) becomes int

- So the compiler gives a type error.
- To fix this, we need to type **cast** (convert it back to byte).

```
byte a = (byte) (6 * 2);
```

↖ value (int) is converted to byte

② Rule : When result type is larger

- If one value is of a larger type (like int, long, float, double), the entire expression becomes that larger type.

```
→ double d = 6 + i + l + f - d;
```

↑ ↑ ↑ ↑ ↑
 int int long float double

Since one operand is **double**, the whole expression becomes **double**.

Some Examples of Type Promotion

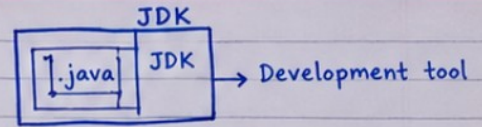
Expression	Result Type (after promotion)
byte + int	int
short + int	int
char + int	int
long + int	long
float + int	float
double + anything	double

③ Example : char operation

```
char a = 'a';
char b = 'b';
char c = a - b; // Error
```

- When we do a - b, both a and b (char) are converted to **int**.
- So the result is an **int**, and we cannot directly store it in a **char** variable.

How is our Code Running ?



① JDK (Java Development Kit)

- It is a development tool for Java.
- Contains **JRE (Java Runtime Environment)** and other tools like **javac** (compiler), **java** (to run program) etc.
- It helps us to **write, compile and run** Java programs.

② JRE (Java Runtime Environment)

- Contains **JVM and libraries** needed to run Java code.
- We need JRE to **run** Java programs.

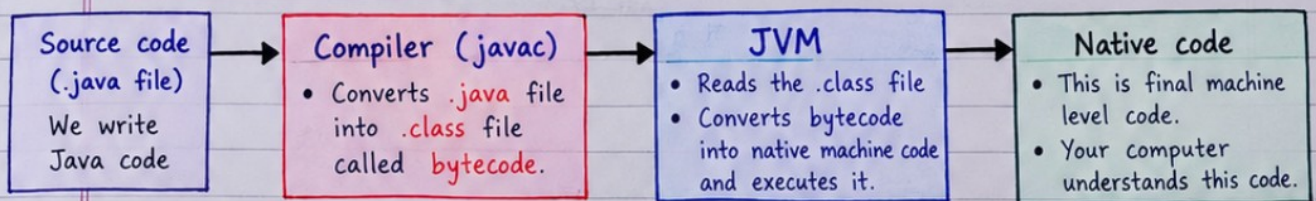
JVM (Java Virtual Machine)

- Runs the **.class** files.
- Converts **bytecode** into **native machine code**.
- Works on different OS (Windows, Mac, Linux)
- Provides **platform independence**.

③ Compiler (javac)

- Compiler (**javac**) converts our **.java** source code into **.class** file (**bytecode**).

Flow of Execution



Detailed Flow

- ① We write code in a **.java** file (source code).
- ② Using **javac** (compiler), it converts **.java** into **.class** file (**bytecode**).
- ③ JVM reads the **.class** file.
- ④ JVM converts **bytecode** into **native machine code** (machine language)
- ⑤ **CPU executes** the native code and the program runs.

Compilation vs Execution

Point	Compilation (javac)	Execution (JVM)
What it does	Converts .java → .class (bytecode)	Runs the .class file and converts bytecode → native machine code
Tool	Compiler (javac)	JVM (inside JRE)
Input file	.java file	.class file (bytecode)
Output	.class file (bytecode)	Runs program (native machine code)
When it happens	Before running	While running

Why Java is Portable ?

- Java runs on any Operating System (Windows, Mac, Linux, etc.).
- Because **JVM** is available on every OS.
- We just need to install JRE/JDK and the same **.class** (bytecode) can run without changing the code.
- That is why Java is **platform independent**.